



Topics covered :

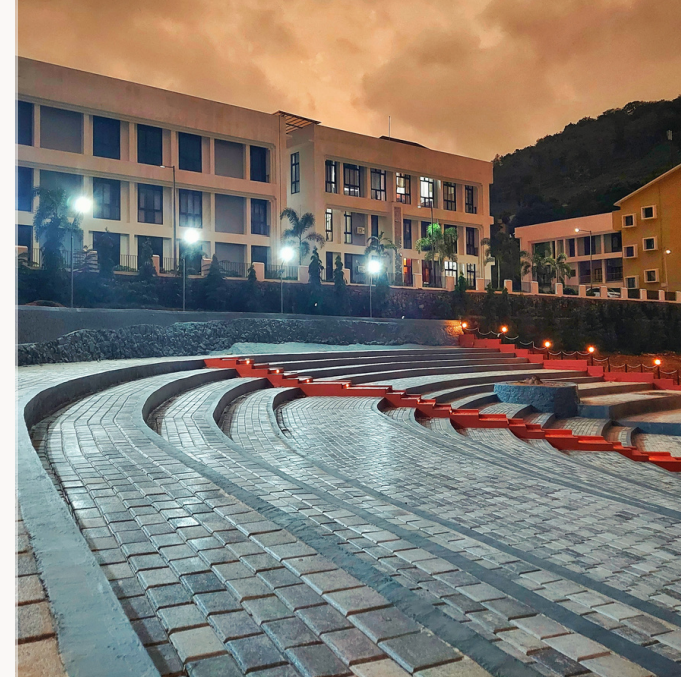
- Introduction to Optimization and its significance in AI
- Fundamentals of Machine Learning
- Deep Learning and Gradient Optimization
- Constrained Optimization in Machine Learning
- Combinatorial Optimization in Machine Learning and AI
- Stochastic Optimization Algorithms in Reinforcement Learning
- Bayesian Optimization and Machine Learning
- Multi-objective Optimization
- Introduction to Time Series and Forecasting
- Evolutionary computation: Functionality and applications

Objectives of the workshop

- 1.To gain insights into the importance of optimization for decision making in artificial intelligence.
- 2.To learn about various optimization techniques and their practical applications in the real-world scenarios.
- 3.To explore advanced optimization methods that leverage probabilistic models and statistical inference.
- 4.To explore research opportunities and challenges in the fields of AI and optimization, fostering discussions on emerging trends and future directions.

About the workshop

Join us for an insightful workshop on **Amplifying AI: Advanced Optimization Strategies for Enhanced Performance (hybrid mode)**. This workshop delves into advanced strategies for optimizing artificial intelligence models to achieve maximum performance across various applications. From hyper-parameter tuning in deep learning to finding optimal policies in reinforcement learning, solving complex combinatorial optimization problems, and solving expensive black-box optimization problems, this workshop covers a wide range of topics to enhance decision-making processes in AI.



Indian Institute of
Information Technology
Kottayam

Five day workshop cum FDP on

**AMPLIFYING AI: ADVANCED
OPTIMIZATION STRATEGIES FOR
ENHANCED PERFORMANCE
(HYBRID MODE)**

MAY 27-31, 2024

**Department of Computational Science and
Humanities**

www.iiitkottayam.ac.in

ABOUT THE INSTITUTE

The Indian Institute of Information Technology, Kottayam is one among the IIITs that has been established as "Institutions of National Importance" by the Ministry of Education, Govt. of India under the ambit of IIIT (PPP) Act 2017. The institute offers 4 year B.Tech and B.Tech(Hons) program with specialized courses covering all major research domains of Computer Science, Electronics and Communication Engineering. The students are admitted based on the JEE main rank list and the JoSAA/CSAB counseling. The institute also offers MTech in CSE and Ph.D. programs in CSE, ECE and Mathematics.

CONTACT

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ABOUT THE DEPARTMENT

The Department of Computational Science and Humanities was formally established in 2019 with an aim to develop mathematically sound young minds for solving industry-related problems arising from different industries and related areas. It offers core courses in Mathematics/Statistics/Humanities to all B.Tech., M.Tech, and Ph.D. programs. This department takes pride in offering an exclusive doctoral program in Mathematics, meticulously crafted to meet the requirements of highly motivated and aspiring graduates holding a degree in this field.

RESOURCE PERSONS

- Dr. Sriparna Saha, IIT Patna
- Dr. Srijith P.K., IIT Hyderabad
- Dr. Pandiri Venkatesh, IIIT Kancheepuram
- Dr. Lakshmanan Kailasam, IIT BHU
- Dr. Pritam Anand, DAIICT, Gandhinagar
- Dr. Jayakrushna Sahoo, IIIT Kottayam
- Dr. Susheel Kumar Joshi, IIIT Kottayam
- Dr. P.Victor Paul, IIIT Kottayam
- Dr. Cinu C Killoor, IIIT Kottayam
- Dr. John Paul Martin, IIIT Kottayam
- Dr. Rubell Marion Lincy, IIIT Kottayam

PATRONS

- Prof. (Dr.) Rajiv Dharaskar, Director, IIIT Kottayam.
- Dr. M.Radhakrishnan, Registrar, IIIT Kottayam.
- Prof. (Dr.) P. Mohanan, Prof. In-Charge (Academics), IIIT Kottayam.

PROGRAM COORDINATORS

- Dr. Susheel Kumar Joshi, Assistant Professor
 - Dr. Dhanyamol M V, Assistant Professor and Head
 - Dr. T. Bakkyaraj, Assistant Professor
 - Dr. Murugan D, Assistant Professor
- Department of Computational Science and Humanities,
IIIT Kottayam*

FEE PAYMENT DETAILS

E-payment link: <https://tinyurl.com/bdhrvvf8>

Steps for payment

1. Choose the category as Educational Institutions
2. Choose payee as IIIT KOTTAYAM
3. Choose payment category as FDP on Amplifying AI-Advanced Optimization

REGISTRATION

Target Audience: Faculty, PhD scholars and UG/PG students.

Registration deadline: 22-05-2024

Registration Link: <https://forms.gle/VLBYYokWdHs44rW4X8>
(Seats will be assigned on a first-come, first-served basis due to limited availability)

REGISTRATION FEE

Participant type	Online	Offline*
PhD/PG/UG	1000/-	2000/-
Faculty	2000/-	3000/-

***Only working lunch and tea will be provided to the offline participants.**

Please note : Registration fee is non refundable.

ACCOMODATION FEE DETAILS

Single room : ₹200/- per day

Shared room: ₹100/- per day

[Provided upon request via mail and mentioned the same in the registration link.]



SCAN FOR REGISTRATION